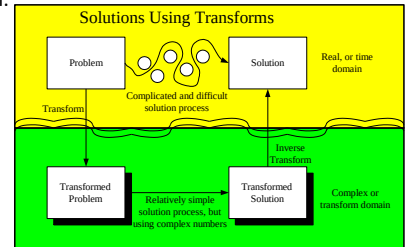


Chapter 5 (additional slides)

AC Circuit Analysis Using Transforms

- The fundamental idea about phasor analysis is that circuits that have sinusoidal sources can be solved much more easily if we use a technique called transformation.

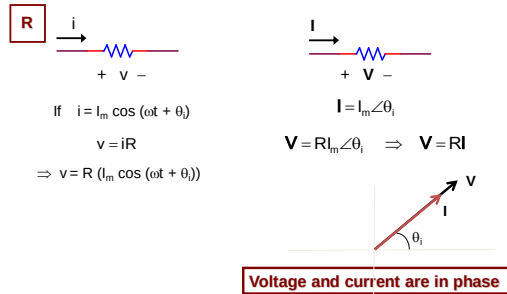


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- In a transform solution, we transform the problem into another form. Once transformed, The solution process uses complex numbers, but is otherwise straightforward.
- The solution obtained is a transformed solution, which must then be inverse transformed to get the answer.
- It is surprising that a process that uses three steps is faster and easier than a process that uses one step, but the steps are so much easier, it is still true.

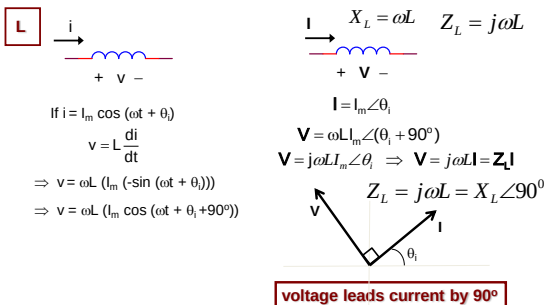
Phasor Relationships for R, L and C

The relationships between V and I for R , L and C are needed in order for us to do the AC circuit analysis



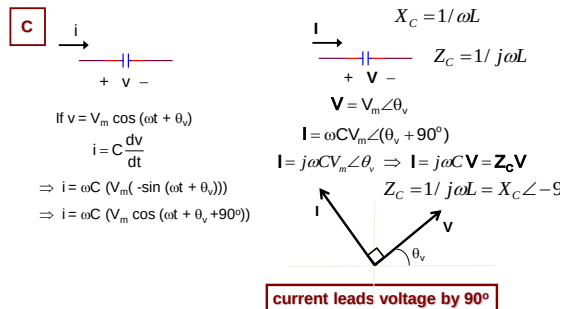
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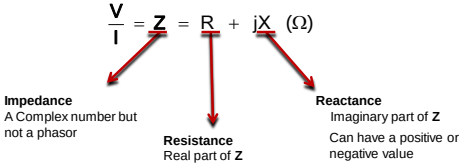
Phasor Relationships for R, L and C

The relationships between V and I for R , L and C are needed in order for us to do the AC circuit analysis



Impedance

The ratio of voltage phasor and current phasor is known as **IMPEDANCE**



Impedance

The ratio of voltage phasor and current phasor is known as **IMPEDANCE**

$$\frac{V}{I} = Z = R + jX \ (\Omega)$$

If $V = V_m \angle \theta_v$ and $I = I_m \angle \theta_i$, then

$$Z = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle (\theta_v - \theta_i) = Z \angle \theta_z$$

and,

$$Z = \sqrt{R^2 + X^2} \quad , \quad \theta_z = \tan^{-1} \frac{X}{R}$$

Impedance

The ratio of voltage phasor and current phasor is known as **IMPEDANCE**

$$\frac{V}{I} = Z = R + jX \ (\Omega)$$

In frequency domain (or phasor) the relationship between V and I for the three circuit elements (R, L and C) is linear:

$$\frac{V}{I} = R \quad \frac{V}{I} = j\omega L \quad \frac{V}{I} = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$$

Elements	Impedance	Reactance	Resistance
Resistor	R	0	R
Inductor	$j\omega L$	ωL	0
Capacitor	$-j \frac{1}{\omega C}$	$-\frac{1}{\omega C}$	0

- The phasor transforms can be summarized in the table given here. In general, voltages transform to voltage phasors, currents to current phasors, and passive elements to their impedances.

Component	Time Domain Quantity	Phasor Domain Quantity
Voltages	$v_x(t) = V_m \cos(\omega t + \phi_v)$	$V_{xm}(\omega) = V_m \angle \phi_v$
Currents	$i_x(t) = I_m \cos(\omega t + \phi_i)$	$I_{xm}(\omega) = I_m \angle \phi_i$
Resistors	R_x	$Z_{R_x} = R_x$
Inductors	L_x	$Z_{L_x} = j\omega L_x$
Capacitors	C_x	$Z_{C_x} = \frac{1}{j\omega C_x}$

AVERAGE POWER AND POWER FACTOR

- If we take the general case depicted in Fig. and use the following for v and i: $v = V_m \sin(\omega t + \theta_v)$

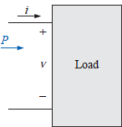
$$i = I_m \sin(\omega t + \theta_i)$$

- then the power is defined by

$$p = vi = V_m \sin(\omega t + \theta_v) I_m \sin(\omega t + \theta_i) = V_m I_m \sin(\omega t + \theta_v) \sin(\omega t + \theta_i)$$

- Using the trigonometric identity

$$\sin A \sin B = \frac{\cos(A - B) - \cos(A + B)}{2}$$



Cont'd

$$\begin{aligned} & \sin(\omega t + \theta_v) \sin(\omega t + \theta_i) \\ &= \frac{\cos[(\omega t + \theta_v) - (\omega t + \theta_i)] - \cos[(\omega t + \theta_v) + (\omega t + \theta_i)]}{2} \\ &= \frac{\cos(\theta_v - \theta_i) - \cos(2\omega t + \theta_v + \theta_i)}{2} \end{aligned}$$

- so that

$$p = \left[\frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \right] - \left[\frac{V_m I_m}{2} \cos(2\omega t + \theta_v + \theta_i) \right]$$

Cont'd

- The first term in the preceding equation, however, has a constant magnitude (no time dependence) and therefore provides some net transfer of energy. This term is referred to as the **average power**.
- The average power, or **real power** as it is sometimes called, is the power delivered to and dissipated by the load. It corresponds to the power calculations performed for dc networks.
- The angle $(\theta_v - \theta_i)$ is the phase angle between v and i . Since $\cos(-\alpha) = \cos\alpha$,

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Cont'd

- the magnitude of average power delivered is independent of whether v leads i or i leads v .
- Defining θ as equal to $|\theta_v - \theta_i|$, where $||$ indicates that only the magnitude is important and the sign is immaterial, we have

$$P = \frac{V_m I_m}{2} \cos \theta \quad (\text{watts, W})$$

$$P = \left(\frac{V_m}{\sqrt{2}} \right) \left(\frac{I_m}{\sqrt{2}} \right) \cos \theta$$

- Since $V_{\text{eff}} = \frac{V_m}{\sqrt{2}}$ and $I_{\text{eff}} = \frac{I_m}{\sqrt{2}}$

$$P = V_{\text{eff}} I_{\text{eff}} \cos \theta$$

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Power Factor

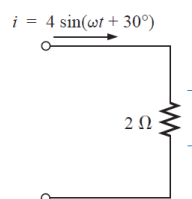
- In the equation $P = (V_m I_m / 2) \cos \theta$, the factor that has significant control over the delivered power level is the $\cos \theta$.
- No matter how large the voltage or current, if $\cos \theta = 0$, the power is zero; if $\cos \theta = 1$, the power delivered is a maximum. Since it has such control, the expression was given the name **power factor** and is defined by

$$\text{Power factor} = F_p = \cos \theta$$

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Example:

- Using complex algebra, find the voltage v for the circuit of Fig. below. Sketch the waveforms of v and i .



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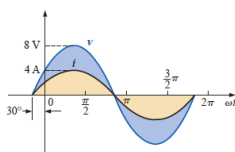
Solution

$$i = 4 \sin(\omega t + 30^\circ) \Rightarrow \text{phasor form } \mathbf{I} = 2.828 \text{ A } \angle 30^\circ$$

$$\mathbf{V} = \mathbf{I} \mathbf{Z}_R = (I \angle \theta)(R \angle 0^\circ) = (2.828 \text{ A } \angle 30^\circ)(2 \Omega \angle 0^\circ)$$

$$= 5.656 \text{ V } \angle 30^\circ$$

and $v = \sqrt{2}(5.656) \sin(\omega t + 30^\circ) = 8.0 \sin(\omega t + 30^\circ)$

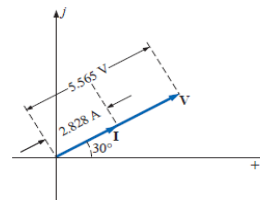


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- It is often helpful in the analysis of networks to have a **phasor diagram**, which shows at a glance the magnitudes and phase relations among the various quantities within the network.

Fig. Phasor diagram for the previous Example



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KVL and KCL in AC circuit

Exactly the same laws (as we have seen in DC circuit) apply for AC circuit:

However, in AC circuit we will deal with **complex numbers**.

KVL: Sum of voltages around a closed loop = 0

$$v_1(t) + v_2(t) + v_3(t) + \dots + v_n(t) = 0$$

$$\rightarrow \mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 + \dots + \mathbf{V}_n = 0$$

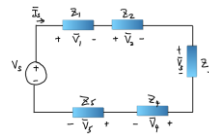
KCL: Sum of currents leaving (or entering) a node = 0

$$i_1(t) + i_2(t) + i_3(t) + \dots + i_n(t) = 0$$

$$\rightarrow \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3 + \dots + \mathbf{I}_n = 0$$

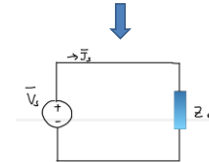
Voltage-division and Current-division rules

Series connected impedances:



$$\mathbf{V}_s = \mathbf{I}(\mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_4 + \mathbf{Z}_5)$$

$$\Rightarrow \frac{\mathbf{V}_1}{\mathbf{I}_s} = \mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_4 + \mathbf{Z}_5$$

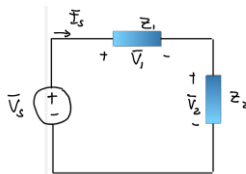


$$\mathbf{V}_1 = \mathbf{I}(\mathbf{Z}_{eq})$$

$$\Rightarrow \mathbf{Z}_{eq} = \mathbf{Z}_1 + \mathbf{Z}_2 + \mathbf{Z}_3 + \mathbf{Z}_4 + \mathbf{Z}_5$$

Voltage-division and Current-division rules

Voltage-division rule:

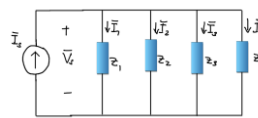


$$\mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}_1 + \mathbf{Z}_2}, \quad \mathbf{V}_2 = \mathbf{I}_s \mathbf{Z}_2$$

$$\therefore \mathbf{V}_2 = \left(\frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \right) \mathbf{V}_s$$

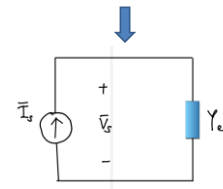
Voltage-division and Current-division rules

Parallel connected impedances:



$$\mathbf{I}_s = \mathbf{V}(\mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4)$$

$$\Rightarrow \frac{\mathbf{I}_s}{\mathbf{V}_s} = \mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4$$

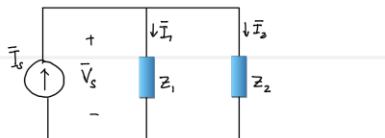


$$\mathbf{I}_s = \mathbf{V}(\mathbf{Y}_{eq})$$

$$\Rightarrow \mathbf{Y}_{eq} = \mathbf{Y}_1 + \mathbf{Y}_2 + \mathbf{Y}_3 + \mathbf{Y}_4$$

Voltage-division and Current-division rules

Current-division rule:



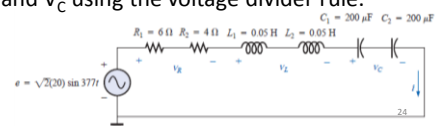
$$\mathbf{V}_s = \mathbf{I}_s \left(\frac{\mathbf{Z}_1 \mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \right), \quad \mathbf{I}_1 = \frac{\mathbf{V}_s}{\mathbf{Z}_1}$$

$$\therefore \mathbf{I}_1 = \left(\frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \right) \mathbf{I}_s$$

Example

• For the circuit of Fig. below:

- Calculate $\mathbf{I}$, $\mathbf{V}_R$, $\mathbf{V}_L$, and $\mathbf{V}_C$ in phasor form.
- Calculate the total power factor.
- Calculate the average power delivered to the circuit.
- Draw the phasor diagram.
- Obtain the phasor sum of $\mathbf{V}_R$, $\mathbf{V}_L$, and $\mathbf{V}_C$, and show that it equals the input voltage $\mathbf{E}$.
- Find $\mathbf{V}_R$ and $\mathbf{V}_C$ using the voltage divider rule.

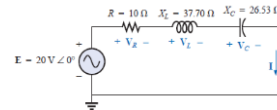


Solution

a) Combining common elements and finding the reactance of the inductor and capacitor, we obtain

$$\begin{aligned}
 R_T &= 6 \, \Omega + 4 \, \Omega = 10 \, \Omega \\
 L_T &= 0.05 \, \text{H} + 0.05 \, \text{H} = 0.1 \, \text{H} \\
 C_T &= \frac{200 \, \mu\text{F}}{2} = 100 \, \mu\text{F} \\
 X_L &= \omega L = (377 \, \text{rad/s})(0.1 \, \text{H}) = 37.70 \, \Omega \\
 X_C &= \frac{1}{\omega C} = \frac{1}{(377 \, \text{rad/s})(100 \times 10^{-6} \, \text{F})} = \frac{10^6 \, \Omega}{37,700} = 26.53 \, \Omega
 \end{aligned}$$

• Redraw the circuit using phasor notation:



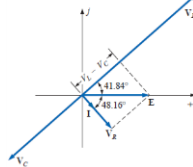
- For this circuit: $Z_T = R \angle 0^\circ + X_L \angle 90^\circ + X_C \angle -90^\circ$
 $= 10 \, \Omega + j 37.70 \, \Omega - j 26.53 \, \Omega$
 $= 10 \, \Omega + j 11.17 \, \Omega = 15 \, \Omega \angle 48.16^\circ$
- The current I is $I = \frac{E}{Z_T} = \frac{20 \, \text{V} \angle 0^\circ}{15 \, \Omega \angle 48.16^\circ} = 1.33 \, \text{A} \angle -48.16^\circ$
- The voltage across the resistor, inductor, and capacitor can be found using Ohm's law:
 $V_R = I Z_R = (I \angle \theta)(R \angle 0^\circ) = (1.33 \, \text{A} \angle -48.16^\circ)(10 \, \Omega \angle 0^\circ)$
 $= 13.30 \, \text{V} \angle -48.16^\circ$
 $V_L = I Z_L = (I \angle \theta)(X_L \angle 90^\circ) = (1.33 \, \text{A} \angle -48.16^\circ)(37.70 \, \Omega \angle 90^\circ)$
 $= 50.14 \, \text{V} \angle 41.84^\circ$
 $V_C = I Z_C = (I \angle \theta)(X_C \angle -90^\circ) = (1.33 \, \text{A} \angle -48.16^\circ)(26.53 \, \Omega \angle -90^\circ)$
 $= 35.28 \, \text{V} \angle -138.16^\circ$

b) The total power factor, determined by the angle between the applied voltage E and the resulting current I, is 48.16° :

$$\begin{aligned}
 F_p &= \cos \theta = \cos 48.16^\circ = 0.667 \text{ lagging} \\
 \text{or} \quad F_p &= \cos \theta = \frac{R}{Z_T} = \frac{10 \, \Omega}{15 \, \Omega} = 0.667 \text{ lagging}
 \end{aligned}$$

c) The total power in watts delivered to the circuit is $P_T = EI \cos \theta = (20 \, \text{V})(1.33 \, \text{A})(0.667) = 17.74 \, \text{W}$

d) The phasor diagram



e. The phasor sum of V_R , V_L , and V_C is

$$\begin{aligned}
 E &= V_R + V_L + V_C \\
 &= 13.30 \, \text{V} \angle -48.16^\circ + 50.14 \, \text{V} \angle 41.84^\circ + 35.28 \, \text{V} \angle -138.16^\circ \\
 E &= 13.30 \, \text{V} \angle -48.16^\circ + 14.86 \, \text{V} \angle 41.84^\circ
 \end{aligned}$$

Therefore, $E = \sqrt{(13.30 \, \text{V})^2 + (14.86 \, \text{V})^2} = 20 \, \text{V}$

and $\theta_E = 0^\circ$ (from phasor diagram)

and $E = 20 \angle 0^\circ$

f.

$$\begin{aligned}
 V_R &= \frac{Z_R E}{Z_T} = \frac{(10 \, \Omega \angle 0^\circ)(20 \, \text{V} \angle 0^\circ)}{15 \, \Omega \angle 48.16^\circ} = \frac{200 \, \text{V} \angle 0^\circ}{15 \angle 48.16^\circ} \\
 &= 13.3 \, \text{V} \angle -48.16^\circ \\
 V_C &= \frac{Z_C E}{Z_T} = \frac{(26.5 \, \Omega \angle -90^\circ)(20 \, \text{V} \angle 0^\circ)}{15 \, \Omega \angle 48.16^\circ} = \frac{530.6 \, \text{V} \angle -90^\circ}{15 \angle 48.16^\circ} \\
 &= 35.37 \, \text{V} \angle -138.16^\circ
 \end{aligned}$$

Exercise

- For the circuit of Fig. below:
- a) Find the total impedance Z_T and the admittance Y_T .
- b) Find the source current I_s in phasor form.
- c) Find the currents I_1 and I_2 in phasor form.
- d) Find the voltages V_1 and V_{ab} in phasor form.
- e) Find the average power delivered to the network.
- f) Find the power factor of the network, and indicate whether it is leading or lagging.

